

Face Reconstruction - 3DSMC

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Abstract

This project presents a 3D face reconstruction pipeline based on the analysis by synthesis paradigm. Given an input image or video sequence, we recover the underlying facial geometry, appearance, and expression by fitting the Basel Face Model to the input via energy minimisation. The optimisation is solved in a coarse to fine manner, starting from sparse landmark alignment and progressively introducing dense depth and photometric terms. The pipeline is evaluated on the Pandora face dataset, allowing comparison between RGB and RGB-D modalities. As an extension, we explore expression transfer, where the recovered expression parameters of one subject are applied to the identity of another.

1. Technical Approach

This project focuses on building an analysis-by-synthesis pipeline for 3D Face Reconstruction and Expression Transfer. By leveraging the parametric Basel Face Model (BFM), the system takes raw target data (single images, video sequences, RGB or RGB-D) and numerically recovers the underlying 3D facial geometry, head pose, expression, and illumination parameters.

Parametric model. $M_i(P)$ denotes the i -th vertex of the BFM, parameterised by $P = (\Phi, \alpha, \beta, \delta, \gamma)$: head pose $\Phi = (R, t)$, identity α , albedo β , expression δ , and spherical-harmonics illumination γ . A vertex is projected via $\pi(M_i(P)) = K(R M_i(P) + t)$.

Energy minimization. We recover P by minimizing a sum of sparse and dense data terms together with regularisation terms:

$$E(P) = \underbrace{E_{\text{dense}}(P)}_{\text{geom. + photo.}} + \underbrace{E_{\text{sparse}}(P)}_{\text{landmarks}} + \underbrace{E_{\text{reg}}(P)}_{\text{prior}}. \quad (1)$$

The dense geometric term combines point-to-point (and potentially point-to-plane) residuals against the depth map D , and the photometric term compares the rendered surface

color to the input image I :

$$E_{\text{geom}} = \sum_i \|M_i(P) - D(\pi(M_i(P)))\|_2^2 + [(M_i(P) - D(\pi(M_i(P)))) \cdot n_i]^2, \quad (2)$$

$$E_{\text{col}} = \sum_i \|M_i(P)_c - I(\pi(M_i(P)))\|_2^2. \quad (3)$$

The sparse term anchors the model to detected landmarks in the photo C_k via the BFM $\leftrightarrow$ detector index map $v(k)$:

$$E_{\text{sparse}} = \sum_k \|\pi(M_{v(k)}(P)) - C_k\|_2^2. \quad (4)$$

The regularizer is needed to prevent large deviations from the PCA mean of the BFM:

$$E_{\text{reg}} = \sum_i \left[\left(\frac{\alpha_i}{\sigma_{\text{id}}^2} \right)^2 + \left(\frac{\beta_i}{\sigma_{\text{alb}}^2} \right)^2 \right] + \sum_i \left(\frac{\delta_i}{\sigma_{\text{exp}}^2} \right)^2. \quad (5)$$

Differentiable rendering. The dense photometric term requires gradients of pixel colors w.r.t. P . We use fixed-visibility differentiable rendering: during the forward pass the rasterizer stores per-pixel triangle index and barycentric weights $(\lambda_i, \lambda_j, \lambda_k)$ as auxiliary render targets, so each rendered pixel is a linear combination of three vertex colors,

$$C_{\text{pixel}}(P) = \lambda_i C_{p_i}(P) + \lambda_j C_{p_j}(P) + \lambda_k C_{p_k}(P), \quad (6)$$

whose derivatives propagate analytically through the parametric model:

$$\frac{\partial C_{\text{pixel}}}{\partial P} = \sum_{m \in \{i, j, k\}} \lambda_m \frac{\partial C_{p_m}}{\partial P}. \quad (7)$$

The partial derivatives w.r.t. all model parameters are then used to update P via Levenberg–Marquardt.

Optimization strategy. We minimize Eq. (1) in a two-stage coarse-to-fine schedule. First, $E_{\text{sparse}} + E_{\text{reg}}$ is minimized to obtain a robust pose and coarse identity initialization. We then add the dense terms and jointly refine P over a Gaussian image pyramid: each level is solved to convergence and used to initialise the next finer level, which improves the conditioning of the non-linear least squares and reduces overall runtime,

$$P^* =_P \lambda_p E_{\text{col}} + \lambda_d E_{\text{geom}} + \lambda_s E_{\text{sparse}} + \lambda_r E_{\text{reg}}. \quad (8)$$

065	Expression transfer.	After fitting both a source and a target frame, we exploit the additive, identity-independent expression basis of the BFM to perform expression transfer as a one-line coefficient substitution: $(\alpha_s, \beta_s, \Phi_s, \gamma_s, \delta_t)$. The resulting parameter set is re-rendered to produce a face that preserves the source identity while adopting the target expression. If time permits, we will replace the source albedo β_s with non-human textures (e.g. animal faces) to explore the creative potential of the reconstructed 3D geometry.	112
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074	2. Requirements		
075	Utilized libraries:	HighFive, dlib, Ceres Solver, Eigen, OpenCV.	
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077		The pipeline is implemented in C++ and relies on the following components. As the parametric face representation we use the Basel Face Model 2017 (BFM-2017). The model is distributed as an h5 file, which we load using HighFive an interface for HDF5. For 2D facial landmark detection, required for the sparse loss term, we use dlib. As our dataset we use the Pandora Face dataset. We also aim to use OpenCV for tooling such as gaussian pyramids. As in previous exercises, we utilise Ceres or non-linear least squares optimization, as well as Eigen.	
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087	3. Milestones		
088		Below we define milestones, moving sequentially throughout the reconstruction pipeline. We aim to build a modular, configurable setup that facilitates comparing different data modalities, optimisers and general configurations.	
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092	3.1. MVP and Dataset Preprocessing		
093	<i>Deadline: June 15th</i>		
094		Before committing to the native C++ pipeline, we will develop a rapid Python-based prototype to parse sample data from the Pandora dataset and initialize the Basel Face Model components. This proof-of-concept phase serves to conceptually validate the pipeline logic, ensure correct dimensionality mapping of facial parameters (α, β, δ) , and verify the alignment math before native implementation begins. Then we will define the dataloader and preprocessing pipeline necessary for Pandora, as well as the dataloader to parse and cache the BFM components. We will facilitate configuration toggles to switch between RGB, RGB-D, video, and single-frame inputs. This forms the foundation for subsequent experimentation.	
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107	3.2. Landmark Detection, Rendering pipeline		
108	<i>Deadline: June 22nd</i>		
109		Integrate a 2D facial landmark detector and differentiable rendering pipeline. This facilitates both the sparse landmark term and dense photometric term	
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	3.3. Optimisation: Sparse		113
	<i>Deadline: July 6th</i>		114
		Complete the first stage of the optimisation using only the sparse term. The goal is a stable coarse alignment of the face model to the input, which serves as the initialisation for the full optimisation.	115
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	3.4. Optimisation: Sparse and Dense		118
	<i>Deadline: July 13th</i>		119
		Extend the optimisation to the full energy function including the dense depth and photometric terms, as well as regularisation. This is the core of the reconstruction pipeline and enables detailed recovery of shape, albedo, expression and illumination parameters.	120
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	4. Evaluation and Analysis Plan		125
		After finishing the implementation, we will do the comparison by testing the pipeline configurations across different setups to evaluate performance and trade-offs. We plan on looking at the following aspects to validate our modular design. First, we will compare data modalities by running parallel experiments using RGB-only versus full RGB-D inputs from the Pandora dataset, allowing us to quantitatively evaluate how the absence of explicit depth maps affects the dense photometric optimization loop. Second, leveraging the capabilities of the Ceres Solver, we will perform an optimization study comparing the Gauss-Newton method against the Levenberg-Marquardt algorithm to analyze their respective convergence behavior, efficiency, and stability during energy minimization. Finally, we will validate our calculated head pose parameters against the Pandora ground truth position and rotation data to compute exact error metrics, paired with a qualitative visual assessment of structural fidelity during the expression transfer phase.	126
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